

# Getting Started With Quarto

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# About this resource

This is a Quarto book is for academic staff who want to get started using Quarto. We cover:

- basic installation and building your first Quarto project
- tips for migrating from a LaTeX source
- using Quarto with a focus on lecture note things like: maths, figures, referencing and much more!

## Note

This guide introduces the fundamentals of Quarto using a **Visual Studio Code (VS Code) workflow**.

Alternatively you can also use Quarto through RStudio, the command line and other routes too!

## Feedback?

If you have any feedback or suggestion on how to make this book more useful then please get in touch with [Josh Lim](#) - University of Bath.

In particular I'd welcome:

- Errors or corrections
- Calls for clarification
- Missing info
- Suggestions for improvements/new topics

# 1 Figures

## 💡 Tip

For full details and advanced options, see the [official Quarto Figures documentation](#):

## 1.1 Basic Figure

The example below creates:

- a numbered figure,
- a visible caption,
- alternative text for screen readers,
- a cross-reference target.

```
![Figure caption](images/dice.png){#fig-dice fig-alt="This is descriptive alt text"}  
  
@fig-dice shows four multi-coloured dice.
```

Figure 1.1 shows four multi-coloured dice.

## 💡 Tip

In Quarto, an image becomes a figure whenever it has a caption and appears by itself in a paragraph.

## 1.2 Layout Options

### 1.2.1 Size



Figure 1.1: Figure caption

```
! [Figure caption] (images/dice.png) {#fig-dice fig-alt="This is descriptive alt text" width=200px}
! [Figure caption] (images/dice.png) {#fig-dice fig-alt="This is descriptive alt text" width=200px}
```



Figure 1.2: Figure caption



Figure 1.3: Figure caption

## 1.2.2 Subfigures

Subfigures allow multiple related images to be grouped under a single figure number with individual sub-captions.

```
::: {#fig-dice-layout layout-ncol=2}
! [Figure caption for left image] (images/dice.png) {#fig-dice-left fig-alt="This is descriptive alt text" width=200px}
! [Figure caption for right image] (images/dice.png) {#fig-dice-right fig-alt="This is descriptive alt text" width=200px}

Here is the main caption for the whole figure.
:::
```

Notes:

- Individual image captions become sub-captions.
- The final paragraph becomes the overall figure caption.
- **Blank lines between images and captions are required.**
- You can also create more custom layouts: `{layout-ncol=3}`, `{layout-nrow=2}`



(a) Figure caption for left image



(b) Figure caption for right image

Figure 1.4: Here is the main caption for the whole figure.

### 1.2.3 Two Images Above One Wide Image

#### **i** Note

For brevity in the next few examples, I have just used pseduocode `{{image.png}}` to show where the image should be

```
::: {layout="[[1,1],[1]]"}
{{image1.png}}
{{image2.png}}
{{image3.png}}
:::
```

### 1.2.4 Unequal Widths

```
::: {layout="[[70,30],[100]]"}
{{large.png}}
```

```
{{small.png}}  
  
{{full.png}}  
  
:::
```

### 1.2.5 Adding Space Between Elements

```
::: {layout="[[40,-20,40],[100]]"}  
  
{{left.png}}  
  
{{right.png}}  
  
{{bottom.png}}  
  
:::
```

Negative values create spacing between items.

---

## 1.3 Figures from code

Figures generated from Python, R, or Julia code are **automatically** treated as figures. See [?@sec-code-figs](#) for examples

---

## 1.4 TikZ figures

The Dan MacKinlay [quarto\\_tikz extension](#) compiles each TikZ block using a local TeX engine. PDF output embeds the compiled PDF directly; HTML output also needs a PDF-to-SVG converter. This project uses `dvisvgm` for that conversion.

### 1.4.1 Setting up TikZ for Quarto

Follow [the quarto\\_tikz installation instructions](#): in the terminal `quarto add danmackinlay/quarto_tikz`

#### ! Important

You will need to run `quarto add danmackinlay/quarto_tikz` each time you create a new Quarto project. This is different behaviour from LaTeX and other extensions, which are installed globally. The reason for this is that Quarto extensions are project-specific, and this allows you to have different versions of the same extension in different projects!

Add the following to your yml preamble:

```
filters:  
- tikz
```

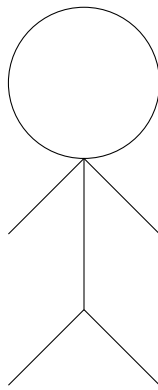


Figure 1.5: A Stick Figure

---

## 1.5 Figure Divs

Any content can be treated as a figure by enclosing it in a fenced div with an identifier beginning with `fig-`. This is useful for:

- embedded videos
- interactive visualisations
- HTML widgets
- custom content that is not a simple image

```
::: {#fig-video}  
  
<iframe width="560" height="315" src="https://www.youtube.com/embed/EbAAmrB0luA?si=jYVDsTRDN  
  
The final paragraph becomes the figure caption.  
  
:::
```

Figure 1.6: The final paragraph becomes the figure caption.